

explain

Q4. How can an element take up the remaining space in a flex container?

- A) flex-grow: 0;
- B) flex-shrink: 0;
- C) flex-basis: 0;
- D) flex: 1;

Q4: How can an element take up the remaining space in a flex container?

✅ **Correct answer: D)** `flex: 1;`

Because `flex: 1` tells a flex item: "grow and take the remaining free space."

First, understand the idea

When a parent is `display: flex`, the children sit in a row (by default).

If one child has `flex: 1`, it will **expand** and fill the leftover space.

✅ Example (easy)

html

```
<div class="box">
  <div class="left">Left</div>
  <div class="middle">Middle (fill space)</div>
  <div class="right">Right</div>
</div>
```

css

```
.box{
  display: flex;
  border: 2px solid;
}

.left, .right{
  width: 100px;
  border: 2px solid gray;
}

.middle{
  flex: 1;          /* ✅ takes remaining space */
  border: 2px solid red;
}
```

✅ Here:

- `left` = 100px
- `right` = 100px
- `middle` = whatever space is left

Why the other options are NOT correct

A) `flex-grow: 0;`

✗ Means **don't grow at all** (stay at your normal size)

CSS

```
.middle{ flex-grow: 0; }
```

So it will **NOT** take remaining space.

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B) `flex-shrink: 0;`

✗ Means **don't shrink when space is small**

It's about preventing shrinking, not filling extra space.

CSS

```
.middle{ flex-shrink: 0; }
```

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C) `flex-basis: 0;`

⚠ This sets the **starting size** of the item to 0, but it **still won't fill space unless it can grow**.

To make it fill space, you'd need something like:

CSS

```
.middle{  
  flex-basis: 0;  
  flex-grow: 1;  
}
```

So **C** alone is not enough.

✅ What exactly is `flex: 1`?

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`flex: 1` is shorthand for:

CSS

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```
flex-grow: 1;  
flex-shrink: 1;  
flex-basis: 0;
```

Meaning:

- **grow** to take extra space ✓
- **shrink** if needed ✓
- start from 0 so sharing space is clean ✓

If you want, I can show you the same concept with **2 items both flex: 1** (they split space equally).

